

Human-AI Interaction in IXR: Design Considerations from Experts

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Abstract

The integration of artificial intelligence (AI) into extended reality (XR) systems enables new forms of intelligent adaptation. However, to enhance user interaction in XR, human-centered design decision-making is required. Despite increasing interest in intelligent XR (IXR), there is limited expert-driven understanding of how XR designers conceptualize useful intelligent adaptation across different task domains. This paper presents an exploratory qualitative study of XR experts' design considerations for IXR systems. Data was collected through a roundtable session at iXR Workshop (1st Workshop on Intelligent XR: Harnessing AI for Next-Generation XR User Experiences (iXR)) using a structured open-ended questionnaire and moderated group discussions. Fifteen XR experts developed and reflected on three scenario-based use cases representing industrial operation, social communication, and creative ideation. Our analysis synthesizes XR experts' perspectives across IXR feature levels and task domains to support early-stage, domain-specific IXR system design.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Mixed / augmented reality.

Keywords

Extended Reality, Intelligent system, Interaction design, Scenario-based design, Contextual design

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1 Introduction

Extended Reality (XR) technologies are evolving rapidly, with increasing adoption across domains such as industrial training, assembly, maintenance, healthcare, and education [1-5]. In parallel, the integration of artificial intelligence (AI) into XR systems has opened new opportunities to enhance user experience through intelligent inference and adaptation [6-8]. This means that, ideally, an XR system should infer the appropriate contextual cues and deliver the right interface and interaction at the right time. However, not all advances in AI necessarily translate into useful or effective experiences when integrated into XR environments. Designing intelligent XR (IXR) systems therefore requires system design decisions grounded in a human-centered approach as poorly aligned choices may increase cognitive load, reduce trust, or introduce safety and ethical concerns in real-world contexts [9-11]. With the same concern, prior work has developed and evaluated IXR systems in different high-stakes contexts with examples ranging from inspection tasks and construction domains, emergency room and triage procedure, etc. [9, 12, 13]. In fact, design decisions which are effective in one setting may introduce friction or risk in another [14-16]. Similarly, research on co-creative AR systems emphasizes the importance of transparency, shared understanding, and explainability in open-ended creative contexts [17].

Despite growing interest in IXR systems development and evaluations, there is limited exploration, expert-driven understanding of how XR experts conceptualize useful intelligent adaptation during the early-stage design process. Existing research often focuses on technical capabilities or system performance, while offering less insight into the design rationales, trade-offs, and concerns that guide expert decision-making across different task domains and social contexts. In this paper, we look deeper into the design decisions involved in the development of intelligent XR systems from the perspective of XR experts. To do so, we organized a roundtable session at the iXR Workshop (1st Workshop on Intelligent XR: Harnessing AI for Next-Generation XR User Experiences (iXR)). The session was structured around an open-ended questionnaire designed to elicit expert perspectives on intelligent adaptation, multimodal interaction, task specificity, and ethical considerations in XR systems. Our research questions in this paper include:

RQ1: What design decisions do XR experts prioritize when designing IXR systems' features?

RQ2: How does task domain influence experts' design choices for IXR systems?

RQ3: What potential pitfalls do XR experts anticipate in deploying IXR systems in real-world settings?

By analyzing the data collected, the primary outcome of this paper is an exploratory, expert-driven understanding of how useful intelligent adaptation in XR systems is conceptualized across diverse XR task domains. The contributions aim to inform human-centered design decisions for IXR systems and support more deliberate integration of AI into XR experiences.

2 Method

2.1 Study Approach

This study adopts an exploratory qualitative approach grounded in expert elicitation to investigate how XR experts conceptualize useful intelligent adaptation in XR systems. Expert elicitation was selected as an appropriate method given the early-stage, design-oriented nature of this inquiry. Rather than evaluating a specific system or testing predefined hypotheses, the goal of this study is to collect expert perspectives, design rationales, and anticipated challenges related to the integration of AI into XR systems across task domains in scenario-based narrations.

The iXR workshop was hosted as part of the 2024 IEEE ISMAR (International Symposium on Mixed and Augmented Reality) conference and was organized by the authors. The roundtable session engaged attendees in a structured open-ended questionnaire designed to support collaborative ideation, discussion of design considerations, and presentation of findings across groups.

2.2 Participants

All participants were XR experts attending the iXR Workshop at IEEE ISMAR 2024. A total of 15 individuals participated in the roundtable data collection session (12 men, 3 women). Participants represented expertise in XR across both academia and industry, including research, system design, and the development of IXR applications. Participation in the roundtable session was voluntary. Responses were collected anonymously without personally identifiable information. The study protocol was reviewed by the Virginia Tech Institutional Review Board and determined to be exempt. For the roundtable activity, participants were organized into three groups of five experts. Each group worked collaboratively on a shared structured open-ended questionnaire, allowing for both individual contributions and group-level discussion within each session. Figure 1 shows the roundtable activity setup, where participants worked in three groups.

2.3 Data Collection

We used a mixed qualitative approach including (1) written responses to the structured open-ended questionnaire and (2) real-time notetaking during moderated group discussions conducted in the report-out session. Figure 2 shows the data collection procedure including subcomponents of round one and round two of the open-ended questionnaire.

Participants engaged with a structured open-ended questionnaire organized into two rounds. In Round One (Scenario Ideation), to reduce groupthink, participants began with a brief individual



Figure 1: Schematic Image of Participants formed three groups to perform structured designed roundtable activity.

ideation period (5 minutes) where participants wrote independent use-case ideas before group convergence. Then, they collaboratively converged on a single scenario, defining target users and core tasks. In Round Two (Design Decisions), participants elaborated on design considerations for human–AI interaction in XR systems within the defined scenario (in adherence to the users and Tasks they defined in previous round). Questionnaire prompts focused on key IXR Features Levels including: (1) Display and Interaction design, (2) intelligent system integration, (3) data and real-time adaptations, and (4) potential pitfalls. Participants were encouraged to supplement their written responses with sketches, hand-drawn diagrams, or other artifacts to further articulate their ideas.

The questionnaire was intentionally open-ended to collect their prioritization in their written responses after evaluating and discussing different design decisions verbally in their group.

2.4 Data Analysis

We followed a hybrid qualitative content analysis approach. Pre-defined analytic dimensions from the structured open-ended questionnaire supported a deductive analysis approach possible. Also, as XR experts shaped three main groups three main scenarios were generated and as a result of their work. Each scenario represents a distinct task domain, including Industrial Operation, Social Communication, and Creative Ideation. These scenarios are used for 1) an inductive analysis of the data to find main patterns in each scenario, and 2) treated as comparative design cases, enabling cross-scenario evaluation of how experts prioritize design decisions about intelligent, adaptive XR systems across different task domains.

All scenario responses were analyzed by a single researcher. To enhance analytic reliability, synthesis was conducted in two passes separated by a time gap of four weeks. In the first pass, design elements were extracted and categorized based on the analytic framework. In the second pass, extracted data was revisited, and refined, with attention to consistency and stability of interpretation over time. Within each category, distinct sub-elements (e.g., specific UI features, data types, or risk concerns) were identified inductively from the written scenario descriptions. Because analysis was single-coder and the participant pool came from a single workshop context, findings should be interpreted as contextual and hypothesis-generating.

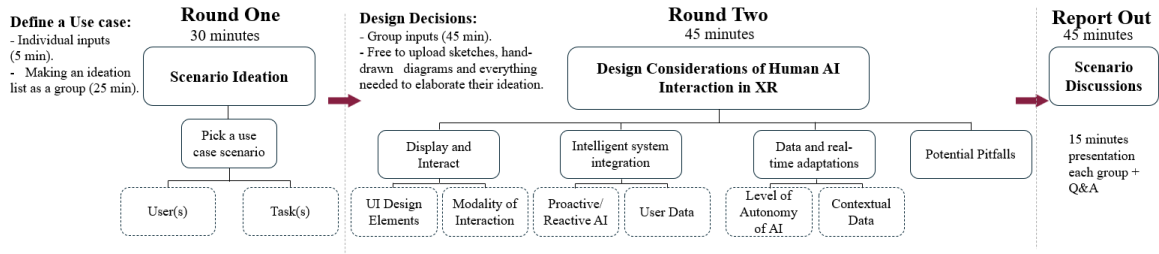


Figure 2: Data Collection Procedure Diagram

3 Results

Participants' responses described IXR systems for three distinct tasks of: 1) Assembly of Complex Object; 2) Visualizing Communication with different modalities; 3) Ever-changing AR decorative items. Table 1 summarizes extracted design elements by scenario (columns) and by IXR feature level (rows), followed by a synthesis column that captures IXR Features Level as well as last row that presents synthesis of Task Domain Level.

3.1 Design Decisions Prioritized in IXR Feature Design

Across all three scenarios, XR experts emphasized natural input modalities, including combinations of gaze, gesture, speech, and direct manipulation. UI elements were designed not only to present information, but also to structure attention, support progress awareness, and enable expressive or corrective user actions depending on context. Across scenarios, interaction design reflected an explicit concern for balancing efficiency with interpretability.

At the intelligent system integration level, experts frequently mentioned human control on adjusting the autonomy level. Proactive behaviors such as task sorting, or summarization were present in all scenarios but were paired with human control and override, including editable AI outputs and user-triggered transitions between automation levels. This indicates a shared design priority toward human-in-the-loop intelligent assistance, rather than considering a fully autonomous system. Finally, experts prioritized user state modeling as a foundational data source type for real-time adaptation. Across scenarios, input data for real-time adaptations was informed by combinations of user performance, confidence, intent, social presence, temporal, and environmental constraints.

3.2 Influence of Task Domain on IXR Design Choices

Design decisions were analyzed and synthesized on the task domain level. To elaborate, in the assembly of complex objects scenario, design choices emphasized task accuracy, safety, and learning support. Interaction elements focused on step-by-step guidance, tool visualization, and progress tracking, while intelligent behaviors were framed as assistive and corrective. Autonomy was acceptable when aligned with user expertise, with systems adapting information granularity and intervention timing based on perceived confidence and performance. The overall interaction emphasis was instrumental, and precision driven.

In contrast, the visualizing communication scenario foregrounded social inclusivity and shared understanding. Here, interaction design centered on speech visualization, thematic clustering, and subtle attention cues, while intelligent behaviors focused on summarization and mediation rather than correction. XR Experts emphasized minimally intrusive autonomy, expressing a preference for human-in-the-loop control to avoid social disruption. Design choices in this domain reflected heightened sensitivity to interpersonal dynamics and interpretive ambiguity.

Decisions for the ever-changing AR decorative items scenario prioritized personalization and emotional regulation. XR Experts envisioned higher degrees of automation driven by temporal, social, and affective context. Interaction was expressive and ambient, and autonomy was conceived as a sliding scale from manual control to full autonomous. This domain expanded IXR design beyond task performance into affective and experiential spaces.

3.3 Anticipated Pitfalls in Real-World IXR Deployment

Across task domains, XR experts identified a range of potential pitfalls associated with each scenario. In assembly domain, experts expressed concerns included over-reliance on intelligent systems, limited model knowledge, and coordination challenges, while in visualizing communication, risks related to social pressure, facial occlusion, and misrepresentation were emphasized, reflecting a tension between social support and individual agency. The decorative AR scenario raised broader concerns around privacy, hardware limitations, data storage, social separation, and personalized content. These risks were not viewed as secondary, but as structurally linked to early-stage design decisions.

4 Discussion

Overall patterns indicate that IXR design decisions are not necessarily transferable across domains, aligning with prior work showing that intelligent XR systems are often designed around domain-specific demands [18-21]. To elaborate, in Industrial Operation setting we noticed that XR experts focused more on design considerations like "task accuracy", "precision", and prioritizing "user control". At the same time, in Social Communication notes, XR experts had more emphasis on design considerations on "Social understanding", "mediating", and "human-in-the-loop" while in Creative Ideation their focus was more on "personalization", "affective",

Table 1: Synthesized data from design considerations of human-AI interaction in XR results

Responses		1) Assembly of Complex Object	2) Visualizing Communication with Different Modalities	3) Ever-changing AR Decorative Items	Synthesis IXR features level
Task Domain		Industrial Operation	Social Communication	Creative Ideation	
Display and Interaction	UI Design Elements	Highlight relevant parts and tools; visualize tool usage; step-by-step AR instructions; collaborative progress visualization	Speech bubbles; thematic clustering of bubbles; color-coded bubbles by speaker/theme; indicator for silent participants; mute indicator	Level-of-detail (LOD) slider; RTS widget; filters; theme selector; visibility toggle; rating system	Direct Control Elements Natural Interaction Progress Awareness Adaptive Level of Details Multimodal system
	Modality of Interaction	Voice commands; gaze tracking; hand tracking	Gaze + gesture (air grab, pin/dart); speech input	Gaze placement; voice description; swipe gestures; walk-up direct manipulation; emotion-based commands (e.g., frown)	feedback & user input
Intelligent System integration	Proactive/Reactive AI	Proactive: task sorting; LOD adjustment based on expertise; personalized UI/modality. Reactive: warnings for errors/danger; repeat steps; adapt procedure if steps skipped	Proactive: automatic summarization; thematic clustering; tangent recognition. Reactive: user edits summarized bubbles	Proactive: calendar-based theme changes; context-based item display; emergency removal; dimmer; mood-based decor; cultural acclimation.	Adjustable Autonomy Human Control Adaptive Assistance
	Level of Autonomy of AI	Initially user-triggered; optional automatic progression	Edit/delete bubbles; clustering assistance	Reactive: user-triggered changes Adjustable autonomy: from manual to fully autonomous; context-guided, sliding autonomy via LOD control	
Data and real-time adaptations	User Data	User confidence; performance; prior experience; frustration level	Gaze; facial expression; speech/voice (speaker distinction)	Calendar; mood; social network (who is present); personal preferences; design style; email	User State Modeling Social Awareness Temporal Sensitivity Personalization
	Contextual Data	Area of interest; user intent; assembly procedure; tool affordances; object tracking; assembly progress; collaboration state	Spatial constraints; background color map (for bubble color coding)	Time/date (holidays); weather; location; space layout; presence/location of others and pets	
Potential Pitfalls		Over-reliance on system; limited model knowledge; merge conflicts in collaboration; privacy concerns	Over-spotlighting introverts; pressure to participate; forced equality; facial occlusion	Privacy loss; dark patterns; hardware limits; over-consumption; social separation; licensing; data storage; lag; eye strain; hyper-capitalism	Control Balance Privacy Risk Social Harm Over-automation Technical Limits Ethical Concerns
Synthesis Task Domain Level		Task accuracy, safety, and learning support Assistive and corrective Instrumental, precision-driven User-triggered first, adaptive with expertise	Social inclusivity and shared understanding Mediating and summarizing Subtle, socially aware augmentation Human-in-the-loop, minimally intrusive	Personalization LOD automation Expressive, affective, ambient Adjustable autonomy from manual to fully autonomous	

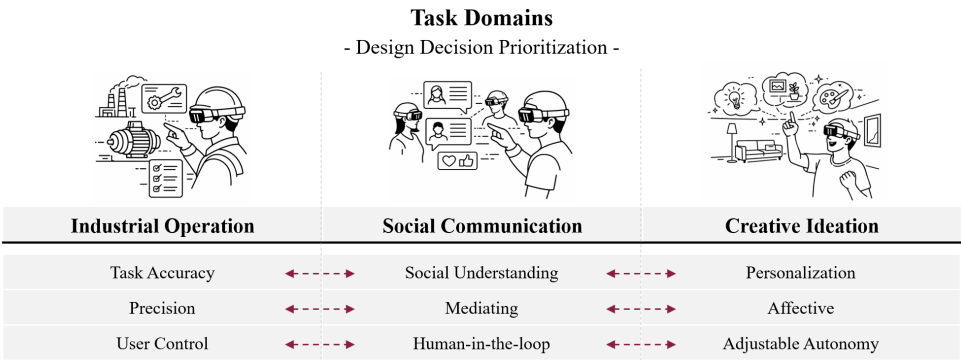


Figure 3: Design Decision Prioritization in the Main Three Task Domains

and “adjustable autonomy”. Figure 3 captures Design Decision Prioritization among the main task domains emerged in this research. Importantly, the prioritizations identified in this work should not be interpreted as prescriptive design recommendations. Rather, they should be understood as early-stage, informed design cues rather than strict rules that must be followed. Notably, this analysis and the associated prioritizations are limited to the task domains represented in the workshop roundtable. Other domains such as education, healthcare, or entertainment entail domain-specific constraints and values that may lead to different design priorities and were not examined in this study [19, 20, 22].

Conclusion

This paper explored how XR experts conceptualize useful intelligent adaptation in IXR systems, based on data collected during a roundtable at the iXR Workshop. Through the analysis of three scenarios representing industrial operation, social communication, and creative ideation, an exploratory synthesis of design considerations was provided. The results indicate different priorities of XR experts across domains and offered educated guidance for initial design decisions that can support more human-centered design of IXR systems.

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